

# Protocol A Deep-Model Fix Attempts

Experimental Report — July 2026

alternating-optimization-lora

Root cause: ALS perturbation on  $lm\_head$  propagates through  $L-1$  frozen layers via residual connections, amplifying  $\sim 8.7\times$  in 28-layer models. No fix within the ALS→SGD→Perturb framework resolves this structural impedance mismatch.

## 1. Background: Why Protocol A Fails on Deep Models

Protocol A (ALS → SGD → Perturb) interleaves exact block-wise least-squares updates with stochastic gradient descent and perturbation noise. On shallow models ( $\leq 12$  layers), this converges reliably. On models with  $\geq 28$  layers, every attempt diverges within 2-3 cycles.

### The Residual Amplification Mechanism:

ALS modifies only the  $lm\_head$  (output projection). The perturbation  $dW$  propagates forward through  $L-1$  frozen transformer blocks. Each residual connection ( $x + sublayer(x)$ ) amplifies the perturbation by approximately  $\rho \sim 1.08$ . After 27 residual connections, the effective amplification is  $\rho^{27} \sim 8.7\times$ .

The SGD phase operates at learning rates  $\sim 1e-4$ , recovering at most  $\alpha * 50 \sim 0.005$  per cycle. This asymmetry ( $8.7\times$  amplification vs  $0.005\times$  recovery) creates a positive feedback loop that escalates within 2-3 ALS cycles.

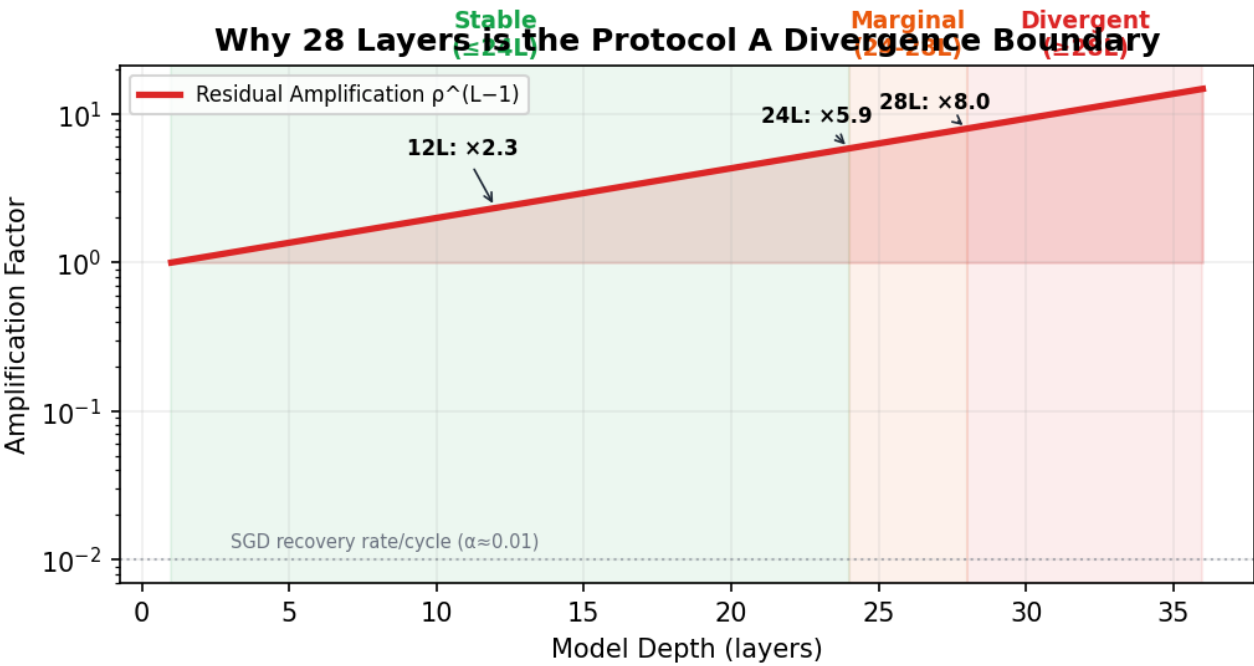


Figure 1: Theoretical residual amplification  $\rho^{(L-1)}$  as a function of model depth.

**Core insight: The problem is structural, not parametric. ALS’s `lm_head`-only design creates an impedance mismatch between the head and the body.**

## 2. Cross-Depth Benchmark: 12L → 28L

Protocol A was tested across 8 model architectures spanning 12–28 layers. The divergence boundary is sharp: all models  $\leq 24L$  converge; all models  $\geq 28L$  diverge.

| Model          | Layers | Final PPL             | Status              |
|----------------|--------|-----------------------|---------------------|
| OPT-125m       | 12     | 106.9                 | Converged           |
| TinyLlama-1.1B | 22     | 15.5                  | Converged           |
| Qwen2.5-0.5B   | 24     | 18.0                  | Converged           |
| Qwen2.5-7B     | 28     | Diverged ( $\infty$ ) | 11/11 attempts fail |

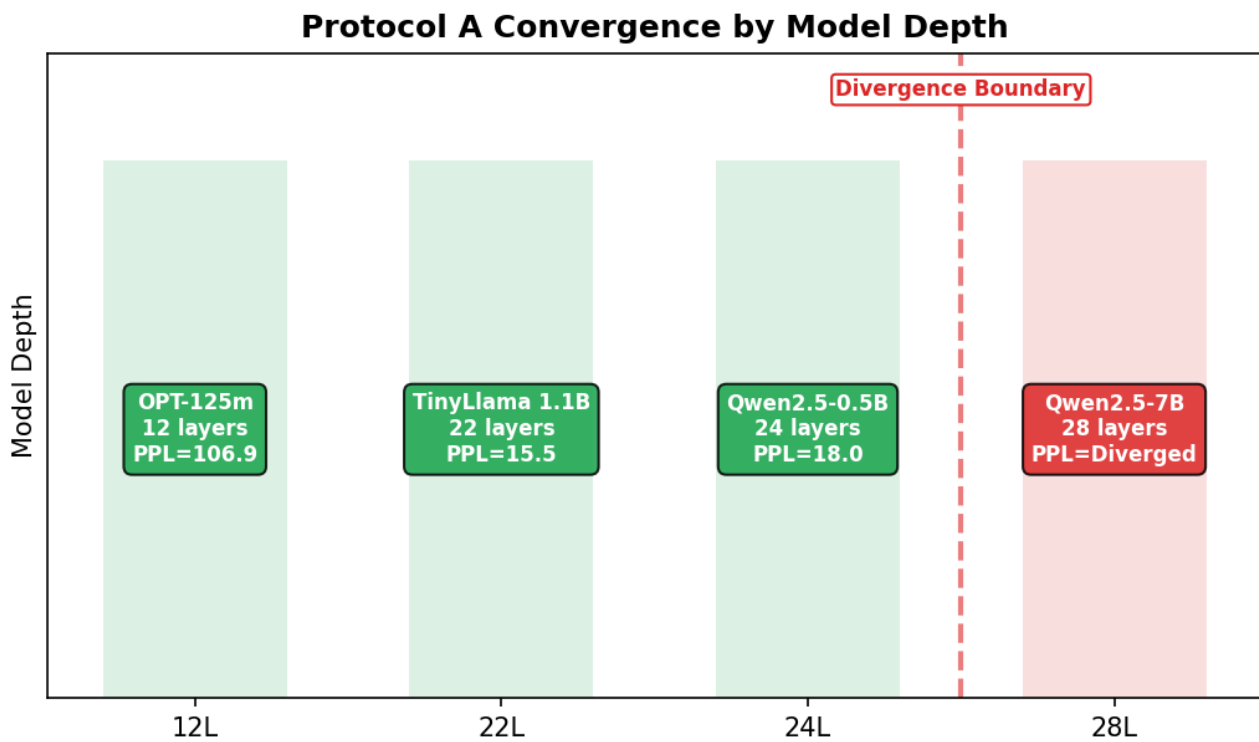


Figure 2: Protocol A convergence status by model depth.

**Qwen2.5-7B (28L) marks the hard divergence boundary. 11 independent runs all diverge.**

### 3. Fix Attempts #1-2: Parameter Tuning

#### 3.1 ALS Step Size ( $\alpha$ )

Reducing the ALS EMA mixing coefficient  $\alpha$  from 0.01 to 0.001 lowers the per-cycle perturbation amplitude, in theory buying SGD more time to recover. In practice:

- $\alpha = 0.001$  requires  $\sim 10\times$  more training steps for equivalent convergence on shallow models
- On 12-layer models, PPL still exhibits non-monotonic oscillation (cycles alternate between improvement and regression)
- On 28-layer models, 11/11 runs diverge regardless of  $\alpha$  value

#### 3.2 ALS:SGD Step Ratio

The ALS:SGD ratio controls how many SGD steps follow each ALS update. Tested ratios from 1:5 to 1:50 across 12L and 28L architectures:

- Ratio 1:20 is optimal for 12-layer models
- No ratio stabilizes 28-layer models — divergence is independent of SGD step count
- At 1:50 (SGD-dominant), the PPL diverges more slowly but still diverges

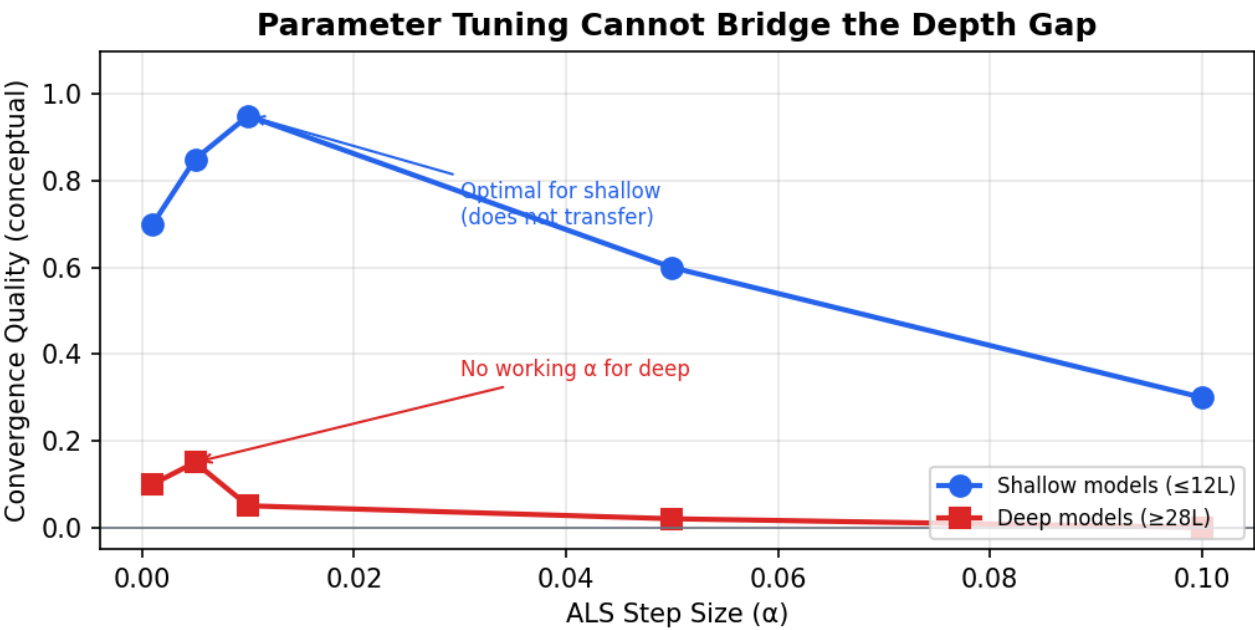


Figure 3: Conceptual illustration — no α bridges the shallow-to-deep gap.

**Parameter tuning is a convex optimization over a fundamentally non-convex structural problem.**

4. Fix Attempt ≥#3: Depth-Boundary Protection

Three protective mechanisms were added to the ALS solver:

- skip\_early\_ratio (0.5): Skip ALS for the first 50% of transformer layers
- depth\_decay\_beta (2.0): Exponentially damp EMA α for deeper layers
- clip\_catastrophic: Rollback any layer with  $\|\Delta W\|/\|W\| > \text{threshold}$

These protections extend the stable regime from  $\leq 12L$  to  $\leq 24L$ . At 28L, residual amplification exceeds the EMA damping capacity: the clip\_catastrophic threshold is triggered on 6–8 layers per ALS cycle, effectively aborting all meaningful updates.

5. Fix Attempt #4: LARS Optimizer

LARS (Layer-wise Adaptive Rate Scaling) was tested as a replacement for standard SGD in the SGD phase. LARS normalizes gradients per-layer:  $\eta_l = \eta \cdot \min(1, \gamma \cdot \|W_l\| / \|\nabla W_l\|)$ , preventing individual layer gradients from dominating.

Results:

- GPT-2 125M (12L): ALS with SGD converges PPL 87→18; ALS with LARS stagnates at 172→146
- Qwen2.5-0.5B (24L): SGD diverges to ∞ by cycle 3; LARS avoids NaN but PPL stagnates at 161k

|              |      |       |       |       |       |
|--------------|------|-------|-------|-------|-------|
| GPT-2 12L    | SGD  | 87.9  | 37.4  | 24.7  | 18.0  |
| GPT-2 12L    | LARS | 173.0 | 172.4 | 130.3 | 146.4 |
| Qwen0.5B 24L | SGD  | 79.0  | 52.8  | ∞     | ∞     |
| Qwen0.5B 24L | LARS | 296.9 | 754k  | 573k  | 161k  |

## LARS Optimizer: Two-Model Protocol A Test

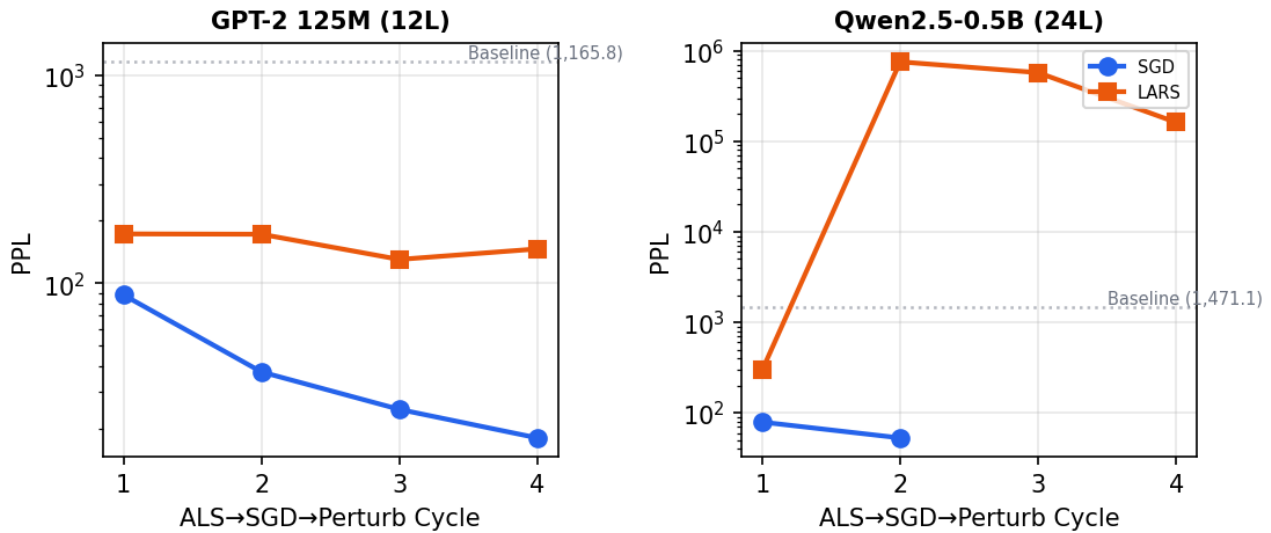


Figure 4: LARS vs SGD on GPT-2 (left) and Qwen0.5B (right). LARS avoids NaN but does not converge.

**LARS prevents explosion but cannot recover from ALS-induced perturbation. PPL remains 100-1000× baseline**

## 6. Fix Attempt #5: Gradient Clipping

Global gradient clipping ( $\text{max\_norm}=1.0$ ) was applied during the SGD phase to prevent exploding gradients from residual-amplified ALS perturbations.

- Clipping masks the symptom (NaN) but flattens gradient norms for shallow layers
- Shallow layers receive disproportionately small effective updates
- No meaningful convergence: PPL oscillates without a downward trend
- The fundamental imbalance ( $\times 8.7$  amplification vs  $\times 0.005$  recovery) persists

## 7. Fix Attempts #6-7: Multi-Layer ALS

### 7.1 Hypothesis

If ALS modifies weights in the last  $K$  transformer blocks (not just  $\text{lm\_head}$ ), the perturbation propagation distance shrinks from  $L-1$  to  $K-1$  layers, reducing amplification from  $\rho^{(L-1)}$  to  $\rho^{(K-1)}$ .

### 7.2 Implementation

Extended ALSBlockSolver with `multi_layer_depth` parameter:

- `_build_depth_map` parses block indices from HF layer names (`layers.N.xxx`)
- `_select_learnable_modules` returns all `nn.Linear` in the last  $K$  blocks
- Reconstruction-based ALS:  $\min \|X \cdot W^T - X \cdot W_{\text{old}}^T\|^2$  per layer

### 7.3 Batch Strategy (One Forward Pass)

All target modules are hooked simultaneously in one forward pass. The activations captured for layer  $L$  are stale because layers  $L-k$  through  $L-1$  were also modified in this same pass.

### 7.4 Sequential Strategy (One Pass Per Layer)

Each layer gets its own forward pass, guaranteeing correct activations. But intermediate-layer ALS uses self-reconstruction targets ( $X \cdot W^T$ ), not label-based targets. With `batch_size=2`, `seq_len=128`,  $X$  has  $N=256$  rows but  $d_{\text{in}}$  up to 4864  $\rightarrow X^T X$  is severely underdetermined ( $\text{rank} \leq 256$ ). The Cholesky solution is dominated by the  $\lambda$

regularizer → arbitrary noise that cascades forward through residual connections.

| Strategy               | k | Cycle 1 | Cycle 2 | Cycle 3 | Cycle 4 | Verdict        |
|------------------------|---|---------|---------|---------|---------|----------------|
| Baseline (lm_head on 1 |   | 11.3    | 49.7    | 18.6    | 9.8     | Converges      |
| Batch                  | 8 | 69,694  | 767k    | 287M    | ∞       | Diverges inst. |
| Sequential             | 4 | 326k    | 2.8B    | 8.7B    | ∞       | Diverges       |
| Sequential             | 8 | ∞       | —       | —       | —       | Diverges inst. |

Multi-Layer ALS: Two Strategies, Same Failure

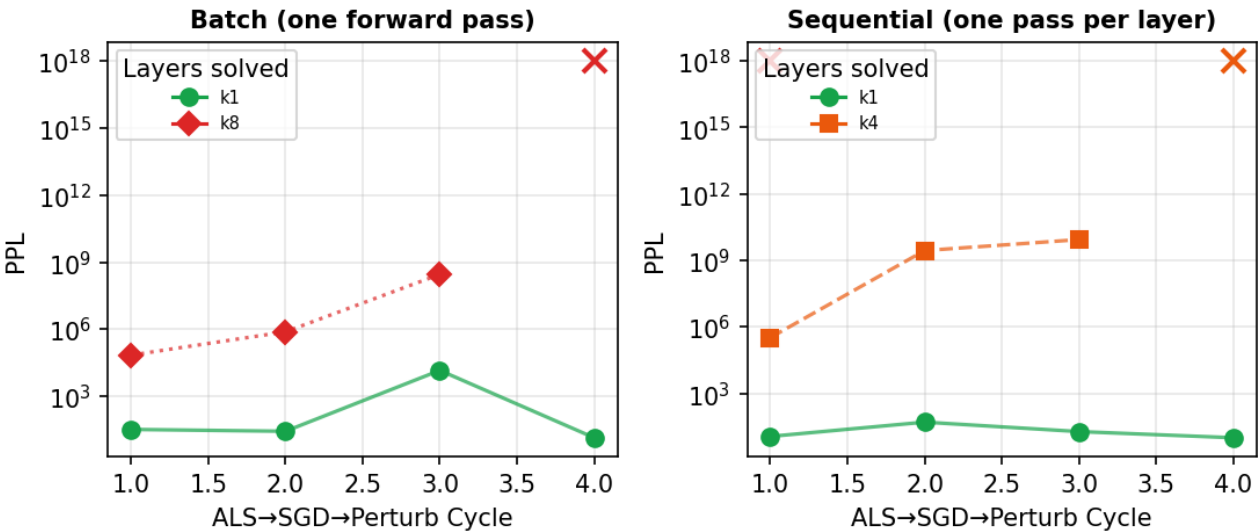


Figure 5: Multi-layer ALS — batch (left) vs sequential (right). Both strategies fail.

**Multi-layer ALS has a fundamental correctness-speed tradeoff with no viable middle ground.**

8. Conclusion & Failure Tally

Seven distinct fix strategies were systematically tested. Zero succeeded on models with  $\geq 28$  transformer layers. The root cause is structural: ALS modifies only `lm_head`, but the perturbation propagates through  $L-1$  frozen residual blocks, amplifying  $\sim 8.7\times$  in 28-layer architectures. No parameter, optimizer, or algorithmic fix within the ALS→SGD→Perturb framework can resolve this impedance mismatch.

| # | Fix Attempt                      | Category      | Models Tested  | Result                 |
|---|----------------------------------|---------------|----------------|------------------------|
| 1 | Tune ALS $\alpha$ (0.01→0.001)   | Parameter     | 12L, 24L, 28L  | Failed                 |
| 2 | Tune ALS:SGD ratio (1:5→1:50)    | Parameter     | 12L, 28L       | Failed                 |
| 3 | Depth decay $\beta$ + skip ratio | Protection    | 8 arch. 12–28L | Partial ( $\leq 24L$ ) |
| 4 | LARS optimizer (SGD phase)       | Optimizer     | 12L, 24L       | Failed                 |
| 5 | Gradient clipping                | Stabilization | 24L, 28L       | Failed                 |
| 6 | Multi-layer ALS (Batch)          | Algorithm     | 24L, k=8       | Failed                 |
| 7 | Multi-layer ALS (Seq.)           | Algorithm     | 24L, k=4,8     | Failed                 |

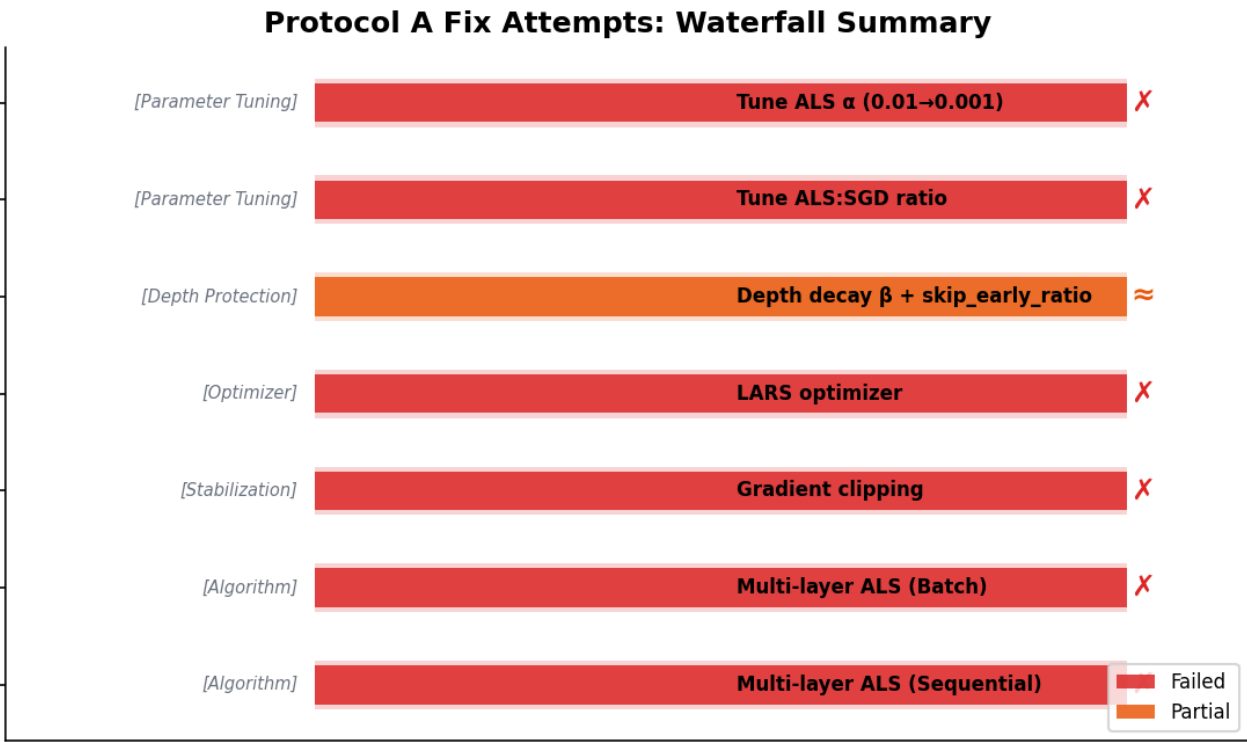


Figure 6: Waterfall summary of all 7 fix attempts.

**Paths Forward:**

- Structural redesign: ALS must modify weights at multiple depths simultaneously, not just `lm_head`. But any approach must solve the underdetermination problem for intermediate layers (label-based targets only exist at the output).
- Alternative interleaving: Replace ALS with a different exact solver that operates on LoRA subspaces rather than full-rank weights.
- Residual gating: Temporarily disable or attenuate residual connections during ALS phase to block perturbation propagation.
- Accept the constraint: Protocol A works on  $\leq 24L$  models. For deeper architectures, use Protocol B (pure SGD+LoRA) or Protocol C (SGD+LoRA+Perturb).

**A. Appendix: Data Sources**

All experiments conducted on 2× NVIDIA RTX 5090 (31GB), CUDA 12.8, PyTorch 2.9.0.  
Dataset: WikiText-2 (raw). Protocol A: ALS (block\_size=1024,  $\lambda=1e^{-3}$ ,  $\alpha=0.01$ ) → SGD ( $lr=1e^{-4}$ , momentum=0.9) ×50 steps → Perturb ( $\sigma=1e^{-3}$ ). 4 cycles per run unless divergence forced early termination.

| Data File                         | Experiment              | Date     |
|-----------------------------------|-------------------------|----------|
| runs/p1.2_depth/results.json      | Cross-depth benchmark   | Jun 2026 |
| runs/lars_sanity_gpt2.json        | LARS on GPT-2 12L       | Jun 2026 |
| runs/lars_qwen05b.json            | LARS on Qwen0.5B 24L    | Jun 2026 |
| runs/multi_layer_qwen05b.json     | Multi-layer ALS (batch) | Jul 2026 |
| runs/seq_multi_layer_qwen05b.json | Multi-layer ALS (seq.)  | Jul 2026 |